

Effects of Respiratory Isolation for Tuberculosis to Reduce Community-based Transmission: A Systematic Review

Ruvandhi R. Nathavitharana,^{1,✉} Abarna Pearl,¹ Amanda Biewer,¹ Laura Young,² Leonard Mukasa,³ Naveed Delrooz,⁴ Advait Subramanian,⁴ Sarah Miller,⁴ Sundari Mase,⁵ Sonal S. Munsiff,⁶ and Edward Nardell⁷

¹Division of Infectious Diseases, Beth Israel Deaconess Medical Center and Harvard Medical School, Boston, Massachusetts, USA; ²Virginia Department of Health, Virginia, USA; ³Arkansas Department of Health, Little Rock, Arkansas, USA; ⁴Tulane University, New Orleans, Louisiana, USA; ⁵Stop TB, Washington, District of Columbia, USA; ⁶Division of Infectious Diseases, University of Rochester School of Medicine and Dentistry, Rochester, New York, USA; and ⁷Division of Global Health Equity, Brigham and Women's Hospital and Harvard Medical School, Boston, Massachusetts, USA

Background. Respiratory isolation of people with pulmonary tuberculosis (TB), including after treatment initiation, is used to prevent community-based transmission; yet guidelines on duration are limited and implementation is heterogeneous. This systematic review synthesized evidence on respiratory isolation for TB to inform National TB Coalition of America guidelines.

Methods. After searching 6 databases, 8 reviewers screened and extracted data in duplicate on effects of respiratory isolation compared to no isolation or masking. Studies were stratified by outcomes: TB infection or disease in contacts, mortality, hospitalization duration, patient and health system costs, and impact on mental health or stigma. We used a convergent integrated approach to synthesize quantitative and qualitative findings and assess limitations.

Results. Seventeen studies were included. There were limited data directly comparing isolation to non-isolation interventions, including effects after treatment initiation. One randomized controlled trial suggested treatment in a sanatorium versus at home did not affect TB incidence in contacts. Modeling studies suggest isolation may reduce transmission but relied on various assumptions, and isolation was implemented alongside other interventions. Descriptive, mixed-methods, and qualitative studies described adverse impacts of isolation on employment, education, food/housing security, and mental health due to transmission fears, stigma, and social isolation. Impacts were compounded in marginalized groups including Indigenous and incarcerated persons.

Conclusions. Data to support current isolation practices, particularly after effective treatment initiation, to reduce TB transmission in communities are limited. Public health guidance should weigh the negative impacts on people with TB against decreased community transmission to make evidence-based decisions about respiratory isolation.

Keywords. tuberculosis; pulmonary tuberculosis; respiratory isolation; isolation; public health.

BACKGROUND

Strengthened efforts to decrease tuberculosis (TB) transmission are urgently needed [1]. Although modeling studies estimate an untreated person with TB disease may infect 5–15 people per year through close contact [2, 3], other cohort studies indicate infectiousness of people with pulmonary TB varies, and many will not lead to secondary infections [4]. Data from diverse experimental studies suggest infectiousness rapidly declines after effective treatment initiation, prior to smear and culture conversion [5–13]; however, the exact duration of treatment to

render people with TB disease non-infectious is uncertain [14]. Respiratory isolation is a widely used public health strategy to reduce TB transmission, yet its evidence base is limited, and recommendations for its duration and implementation are variable [15].

Respiratory isolation is defined as separating people with a respiratory disease from those presumed to be without disease and includes the use of individual rooms/units in institutions, home, or community settings [16]. The most widely implemented recommendation for isolation duration (in the context of health facilities) is 2 weeks following effective (confirmed or presumed) TB treatment initiation [17], but this is longer if smear conversion and clinical improvement are required. This recommendation arose from a 1976 commentary that stated “people with pulmonary TB are likely no longer infectious after 2 weeks on treatment” without specific evidence to support this precise duration, which was derived by consensus [18]. A systematic review that informed the latest World Health Organization (WHO) infection prevention and control (IPC) update (focused on facility-based transmission) in 2019 concluded there was only indirect and/or low-quality evidence for the effectiveness of isolation on TB transmission reduction [16].

Received 30 May 2024; editorial decision 16 September 2024; published online 14 October 2024

Correspondence: R. Nathavitharana, Division of Infectious Diseases, Beth Israel Deaconess Medical Center, 110 Francis St, Ste GB, Boston, MA 02215, USA (rnathavi@bidmc.harvard.edu); E. Nardell, Division of Global Health Equity, Brigham and Women's Hospital, 75 Francis St, Ste GB, Boston, MA 02115, USA (enardell@gmail.com).

Clinical Infectious Diseases® 2025;80(1):189–98

© The Author(s) 2024. Published by Oxford University Press on behalf of Infectious Diseases Society of America. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.
<https://doi.org/10.1093/cid/ciae496>

There is growing recognition of the effects of current TB care standards, including respiratory isolation, on stigmatization and mental health of people with TB [19], with concerns for the human rights implications that isolation restrictions, particularly if prolonged, pose for people affected by TB [20, 21]. Guidance to weigh the potential public health benefits against patient harms of respiratory isolation are lacking [15]. This systematic review synthesizes evidence on the effects of respiratory isolation to prevent community TB transmission on patient- and public health-important outcomes and was commissioned to inform the National TB Coalition of America (NTCA) Guidelines for Respiratory Isolation and Restrictions to Reduce Transmission of Pulmonary Tuberculosis in Community Settings [22].

METHODS

The review protocol was registered on 11 April 2023 on the International Prospective Register of Systematic Reviews (PROSPERO2023 CRD42023393941).

Populations, Interventions, Comparators, and Outcomes (PICO)

Populations were people with pulmonary TB and their contacts. The intervention was respiratory isolation based on any criteria (clinical, microbiological, radiographic), including isolation in a household, community setting, or facility if the purpose of isolation was to prevent community transmission

(rather than facility transmission). The comparators were either no isolation (PICO 1) or masking (PICO 2) (see [Supplementary Materials](#) for PICO question list). Patient and public health important-outcomes were: TB infection among contacts (incidence/prevalence), TB disease among contacts (incidence/prevalence), TB mortality (index patients or contacts), TB hospitalization (duration), patient and health system costs, and effects on mental health and stigma.

Search Strategy

Searches were conducted by a professional librarian in the following databases: MEDLINE (Ovid), Embase (Ovid), Science Citation Index Expanded (Web of Science), Cumulative Index to Nursing and Allied Health Literature (CINAHL, EBSCOHost), Cochrane Central, and WHO Global Index Medicus (final search 3 March 2023) ([Supplementary Materials](#)).

Study Selection

Qualitative, quantitative, and mixed methods study designs were included, without restrictions based on country, setting, or publication date. Systematic reviews were used to find additional publications. We used Covidence [23] to manage study selection. Seven review authors (A. P., A. B., L. Y., L. M., N. D., A. S., and S. M.) screened all titles and abstracts and reviewed full texts using the inclusion and exclusion criteria ([Table 1](#)) in duplicate, with unresolved disagreements reviewed

Table 1. Inclusion and Exclusion Criteria Used During Article Screening Process

Inclusion Criteria		Exclusion Criteria
Types of participants	Studies that include: - Adults and children of all ages - Participant types may include patients, clinicians, clinic managers, programmatic staff, and other stakeholders	- Studies not about tuberculosis - Studies not about the effect of respiratory isolation as defined in our review - Any study that did not report any of the stated outcomes of interest - Any study that is not a primary study but rather an expert commentary (that does not include primary relevant data) or narrative review (systematic reviews and qualitative syntheses were included for full text review)
Types of interventions	Must include: Respiratory isolation (spatial separation) of presumed infectious people with TB (for the purpose of reducing community transmission)	
Types of comparator	- Isolation versus no isolation - Isolation versus masking	
Types of studies	Primary studies (non review/commentary) including: - Randomized trials - Observational studies (cohort, cross-sectional, and case-control); - Pre- and post-studies; - Interrupted time series; - Mixed-methods studies; - Qualitative studies	
Types of outcome measures	- TB infection in contacts - TB disease in contacts - Mortality in patients/contacts - Hospitalization (duration) - Patient and/or health care system costs - Patient mental health and stigma	
Types of settings	- Studies in low-, middle-, or high-income countries at any level of the healthcare system - Studies in hospital, community, and household settings.	

by an independent senior author (R. R. N.). Given the limited data to inform the PICO questions, we included studies if they evaluated the effect of respiratory isolation on the outcomes of interest, even if there was no comparator.

Data Extraction and Quality Assessment

Eight review authors (A. P., A. B., L. Y., L. M., N. D., A. S., S. M., R. R. N.) extracted data independently in duplicate. Details of respiratory isolation including populations, settings, isolation criteria, and comparator interventions were abstracted. Key study findings were summarized narratively using a standardized form. For quantitative studies, crude or adjusted effect measures such as incidence rate ratio or prevalence ratio or odds ratio, as well as covariates used for adjusted analyses, were included. For implementation or mixed-methods studies, intervention rationale and assessment of effectiveness, barriers, and enablers, were included. Hospitalization was measured in days. Costs were measured in US or International dollars [24]. For qualitative studies, data on themes, findings, and supporting quotations were abstracted. If available, data on user values, workflow, resources, implementation, and justice/equity were also abstracted. Conflicts were resolved through consensus with

senior review authors (R. R. N., E. N., S. M., S. S. M.). We undertook quality assessment using the Mixed Methods Assessment Tool (MMAT) [25].

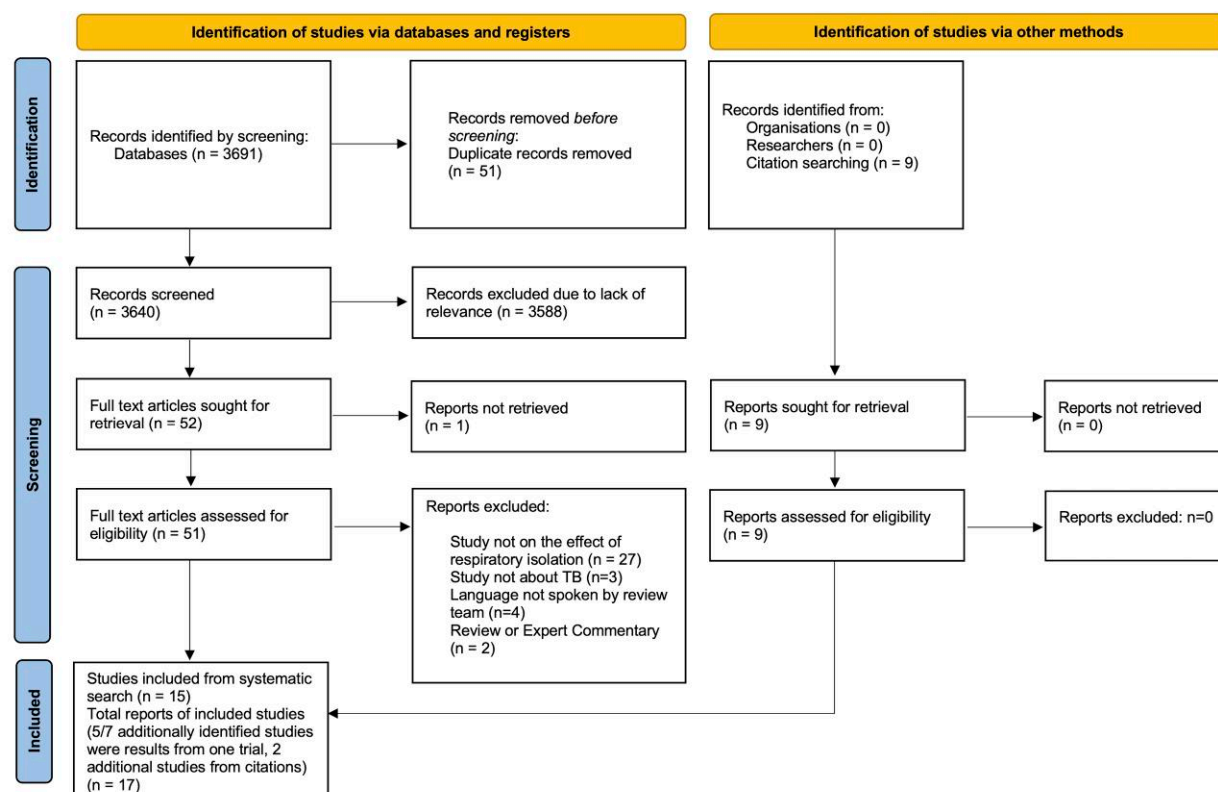
Data Analyses

We used a convergent integrated approach to synthesize quantitative and qualitative findings and assess limitations [26]. Given the heterogeneity of study designs, populations, and outcome measurement approaches, meta-analyses could not be conducted.

RESULTS

Characteristics of Included Studies

The search identified 3691 articles; after removal of duplicates, 3640 abstracts were screened and 60 full-text articles were reviewed (Figure 1). Seventeen articles were included: 1 randomized controlled trial (RCT), 1 quasi-experimental controlled trial, 1 cohort study, 3 modelling studies, 3 quantitative descriptive studies, 1 mixed-methods study, and 7 qualitative studies (Table 2). TB incidence in study countries varied: high-incidence (24%), medium-incidence (18%), low-incidence (47%), and unspecified (model-based studies 12%). Forms of



Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. doi: 10.1136/bmj.n71. For more information, visit: <http://www.prisma-statement.org/>

Figure 1. PRISMA flow diagram showing databases searched; numbers of records identified, sifted, reviewed, and included; and reasons for exclusion. Abbreviation: PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

Table 2. Setting, Design, Population(s) Studied, Intervention(s) Implemented, and Outcome(s) Measured in the Studies Included, Ordered by PICO Question

Study	Setting and Population(s)	Intervention (or Type of Isolation)	Key Findings
PICO 1a, 1b, 1c, 2a, 2b (primary Research Studies): Effect of Respiratory Isolation for TB on TB Infection or Disease in Contacts			
Madras Studies (Andrews [27]; Devadatta 1961; Dawson 1966; Kamat [28]; Ramakrishnan 1961) <i>RCT</i>	Tuberculosis Chemotherapy Centre, Madras (Chennai), India; 193 Index patients (1st family member with pulmonary TB) and 693 family contacts	Newly diagnosed people with smear/culture positive pulmonary TB treated with Isoniazid and PAS were randomized to treatment in a sanatorium versus at home for one year.	• Incident TB disease in contacts at 1 and 5 y was similar: ◦ sanatorium: (22/272 (8%) and 30/272(11%)) ◦ home: (21/256 (8.1%) and 27/256(10.5%)) • Incident TB infection (TST conversion) in contacts at 1 and 5 y was similar: ◦ sanatorium 19/87 (21.8%) and 39/87 (45%) ◦ home: 20/86 (23.3%) and 34/86 (39%) • Major risk was due to TB exposure pre-diagnosis: ◦ 17/26 (65%) contacts developed TB disease within 3 m of index patient exposure. • >80% of contacts (14/17) who developed TB disease were children ≤10 y.
Balcels clinicaltrials.gov /NCT04338596 <i>Quasi-experimental controlled trial</i>	Two healthcare districts in Santiago, Chile (31 clinics); 158 Index patients (and 422 contacts >5 y)	Respiratory bundle (Intervention): N95 respirators for index patient and contacts, separating index patient when possible (isolation), improving household ventilation, TB transmission education versus standard of care (SoC)	• Baseline TB disease in contacts: ◦ Intervention (2.7%) ◦ SoC (2.7%) • No difference in new TB infection (IGRA conversion at 12 wks) in contacts RR 1.05 (0.38–2.86): ◦ Intervention (8.5%) ◦ SoC (8.1%) • Young age trended towards association with increased IGRA conversion (RR_{adj} = 0.96, 95% CI: .92–1.00, <i>P</i> = .057). • There was limited adherence in the intervention arm (43% complete adherence).
Selmane [29] <i>Modeling</i>	Hypothetical (parameters from the literature); Uses four sub-populations: susceptible, latent, infectious, and treated for pulmonary TB	Not clearly defined; some form of hospital isolation	• An optimal control isolation strategy (depends on factors reducing risk of infection) can reduce TB disease and infection respectively by 340–380 people in a population of 6000 over a 6 y period.
Espindola [30] <i>Modeling</i>	Agent-based computer simulation model, using global data, included three disease states: susceptible, latent TB infection, and TB disease	15-day isolation period (location not specified) following start of TB disease treatment	• If 90% PWTB are identified in 4 wks of developing active TB, a 15 d isolation (if 50% effective) period could reduce MDR-TB incidence by 10% over a 10 y period. • 50% isolation + 100% reduced transmissibility outperforms 100% isolation + 50% reduced transmissibility.
Basu [31] <i>Modeling</i>	Rural, high HIV-prevalence region in South Africa; 1300 patients with XDR-TB based on data from KwaZulu-Natal.	Isolation facilities (TB ward in hospital, other hospital areas, trailer park homes; model considered individual isolation vs 5 to 10 person isolation clusters)	• Individual isolation facilities could avert 57% of future XDR-TB. • Staff/respirator use could reduce overall XDR cases by 2% & health worker cases by ~ 66%; surgical mask use by patients could reduce overall cases by 5%. • There was a lower impact of individual (vs combined) IPC measures on XDR-TB transmission.
PICO 1d, 1e: Effect of respiratory isolation for TB on hospitalization or patient or health system costs			
Adler [32] <i>Descriptive</i>	4 Urban hospitals in NYC, USA; 69 patients with culture positive pulmonary TB	Hospital isolation of TB patients (released after 3 negative smears)	• 32 (64%) had prolonged stays (average 30 d), when 2 wks was considered adequate based on treatment initiation and sputum study results. • More than half of those 32 patients were waiting to document 3 negative smears. • At \$450/day, the 32 patients stayed for 536 d and total excess costs were \$262 640.
Oladimeji [33] <i>Descriptive</i>	Urban hospitals in Nigeria; 98 patients with MDR-TB	Voluntary hospital isolation of patients with MDR-TB (fitness for discharge based on after 2 negative sputum results)	• The average hospital stay for patients with MDR-TB was 4.5 ± 1.9 m. • Concern for loss of job and income due to prolonged hospitalization. • Long waiting lists for admission from community.

Table 2. Continued

Study	Setting and Population(s)	Intervention (or Type of Isolation)	Key Findings
Lobue [34] Descriptive	San Diego County, USA; 7 homeless people with pulmonary TB in community housing program	Community isolation location based on infectiousness (low-risk = 3 negative smears)	• Supervised, county-paid housing can save costs (estimated \$27 035 per patient), compared to hospital isolation (cost per patient of \$56/day vs \$651/day).
Rapoport [20] Qualitative	Urban MoH public clinic in Lima, Peru; 85 Adolescents with pulmonary TB; caregivers, HCWs	Voluntary home isolation; instructed to isolate and refrain from activities outside the home for 2 m (until AFB sputum samples from end of first and second month are negative)	• Academic delays caused by isolation affected education and career trajectories, including repeating semesters and risking chances for scholarships. • Loss of employment was also a risk of isolation.
Silva [35] Qualitative	3 public health units and 2 psychiatric hospitals in Toronto area, Canada; 14 health workers and 6 decision-makers re: people with TB & SPMT	Respiratory isolation for first few weeks until deemed non-infectious, could be home versus involuntary isolation	• Patients with SPMT, homelessness, low SES, are disproportionately affected by isolation for TB. • There is a need for social support resources; eg, food vouchers or job security that should not fall on HCWs to provide.
PICO 1f: Effect of respiratory isolation for TB on patient mental health or stigma			
Basham [36] Cohort	Hospitals in British Columbia, Canada; 755 834 immigrants between 1985 and 2015	Hospitalization (not specified whether entirely due to isolation requirements)	• Immigrants with TB had a higher risk of depression (aHR 1.24, 95% CI 1.12–1.31). • Hospital length of stay (hypothesized to represent isolation-induced depression) mediated 50% of risk.
Kelly-Rossini [37] Mixed Methods	Large city hospital in the USA; 18 adults with HIV with AFB-positive smear, under respiratory isolation	Hospital isolation in private room, for 1 or more weeks	• TB patients experienced decreased psychosocial wellbeing, feeling of staff abandonment. • TB patients experienced twice the total mood disturbance compared to smoker controls.
Oladimeji [33] Descriptive	Urban hospitals in Nigeria; 98 people with MDR-TB	Voluntary hospital isolation of people with MDR-TB (released after 2 negative sputum results)	• The impacts of isolation included social disconnection, separation from family and partners, and inability to engage in social/economic activities.
Moffat [38] Qualitative	Public Sanatorium/hospital in Canada; 15 adult First Nations members with personal or family experience of TB	Sanatorium isolation	• Forced isolation in Indigenous patients led to fear of TB recurrence and distrust of health care. • Loss of culture and language during isolation led to social ostracization. • Long absences also led to family disruptions.
Mayan [21] Qualitative	Public clinic in Manitoba Prairie Provinces, Alberta, Canada; 48 Indigenous adults with culture positive TB	Sanatorium isolation	• Negative emotions associated with isolation included feelings of confinement, boredom, and disconnect from others. • Family disconnection, particularly for women, was associated with sadness, guilt, and grief. • Feelings of loss of cultural connection was common.
Rueda [39] Qualitative	Penitentiary center in Medellin, Colombia; 14 incarcerated people	Prison isolation	• Isolation had largely negative impacts ranging from boredom to anger and leading to hostility, aggression, and destructive behaviors.
Silva [35] Qualitative	3 public health units and 2 psychiatric hospitals in Toronto area, Canada; 14 health workers and 6 decision-makers re: people with TB & SPMT	Respiratory isolation for first few weeks until deemed non-infectious, could be home versus involuntary isolation	• Isolation reduced access to important therapeutic activities/interactions. • SPMT could impact adherence to treatment and isolation.
Rapoport [20] Qualitative	Urban MoH public clinic in Lima, Peru; 85 Adolescents with pulmonary TB; caregivers, HCWs	Voluntary home isolation; instructed to isolate and refrain from activities outside the home for 2 m (until AFB sputum samples from end of first & second month are negative)	• Negative psychological effects of isolation included despair from school absence, separation from loved ones, sports exclusion, and experience of anticipated, enacted, and internalized stigma, depression and shame associated with having TB.

Table 2. Continued

Study	Setting and Population(s)	Intervention (or Type of Isolation)	Key Findings
Bedingfield [40] <i>Qualitative</i>	Public clinic in Calgary, Alberta; 6 adults with smear-positive TB with >1 contact and 13 family members	Home isolation	• Fear of transmission was greater than fear of dying (leading to changes in sleeping and eating arrangements). • There was increased stress on the families taking care of patients in home isolation.
Horter [41] <i>Qualitative</i>	A healthcare program in rural northern Uganda; 9 patients with MDR-TB & 2 with mono-resistant TB	Home isolation versus hospital isolation	• Home isolation was preferred to hospital isolation. • Patients worried more about TB transmission in hospitals than at home.

Bold text is to highlight key findings to help the reader.

Abbreviations: CI, confidence interval; HCW, healthcare worker; HIV, human immunodeficiency virus; IGR, interferon-gamma release assay; IPC, infection prevention and control; MDR-TB, multidrug-resistant tuberculosis; MoH, Ministry of Health; PAS, para-aminosalicylic acid (second-line TB drug); PICO, populations, interventions, comparators, and outcomes; PWOTB, people with tuberculosis; RCT, randomized controlled trial; RR, risk ratio; SES, socioeconomic status; SPMI, serious and persistent mental illness; TB, tuberculosis; TST, tuberculin skin test; XDR-TB, extensively drug-resistant tuberculosis.

respiratory isolation included hospital (59%), home (29%), and community/other (24%) ([Supplementary Table 1](#)). Notably, respiratory isolation was rarely implemented as the sole intervention, and only 1 study directly compared isolation versus no isolation. Although all studies evaluated isolation in patients who had been diagnosed and most likely started on treatment, details of treatment including assessment of likely effectiveness were typically not reported.

PICO 1a and 1b: Effect of Respiratory Isolation Compared to No Isolation on TB Infection and Disease in Contacts

A RCT conducted in India in 1956–57 followed 693 close contacts of 193 people with smear- or culture-positive pulmonary TB receiving daily isoniazid and para-aminosalicylic acid (PAS) for 12 months, either isolated in the setting of a sanatorium, or with treatment initiated and continued at home without isolation precautions [42]. Of contacts with a negative baseline tuberculin skin test (TST), 19/87 (21.8%) in the sanatorium arm, and 20/86 (23.3%) in the home arm developed a positive TST at 1 year, and 39/87 (45%) and 34/86 (39%) had TST conversion at 5 years [27, 28]. Incident TB disease at 1 and 5 years was similar for sanatorium (22/272, 8% and 30/272, 11%) and home contacts (21/256, 8.1% and 27/256, 10.5%) [27, 28]. Statistical analyses were not reported. Most contacts who developed TB (17/26, 65%) did so within 3 months of index patient treatment initiation and 81% of these were children under 10 years [27, 28]. Most index patients had moderate-extensive cavitory disease (70% sanatorium, 78% home), and many had heavy bacillary burden on smear (37% sanatorium, 54% home); drug resistance was rare [42].

Three modeling studies suggested isolation (combined with other measures) may reduce transmission [29–31]. One suggested hospital-based isolation could reduce incident TB infection and disease [29]. Another examined early case detection and isolation on multidrug-resistant (MDR) TB epidemiology, suggesting if enough people with TB disease were treated, a 15-day isolation period after treatment initiation could decrease transmission [30]. The third evaluated the effect of infection control measures on community extensively drug-resistant (XDR) TB incidence in South Africa and found individual versus 5-person or 10-person isolation units (isolation duration not specified) could prevent 57%, 37%, and 30% of future XDR-TB cases, respectively [31].

PICO 1c: Effect of Respiratory Isolation Compared to No Isolation on Mortality

The RCT in India found no difference in TB-related mortality amongst index patients treated in a sanatorium versus at home. Three patients died within 12 months (1 sanatorium and 1 home patient who died of TB, and 1 home contact who died of non-tuberculosis causes [27, 42]). For close contacts, over 5 years there were more deaths in the home group (22/256,

8.6%, 1 due to TB) than in the sanatorium group (12/272, 4.4%; $P = .07$, 3 due to TB) [28].

PICO 1d: Effect of Respiratory Isolation Compared to No Isolation on Duration of Hospitalization

A descriptive study of 69 culture-positive TB patients in New York City hospitals found 72% (50/69) of admissions were prolonged (>2 weeks), of which 64% (32/50, average stay of 30 days) were longer than appropriate (lack of medical or non-medical reason for prolonged stay) [32]. In over half, prolonged admissions were due to awaiting negative smears despite clinical improvement, eligibility for discharge, and appropriate home situations. A descriptive study of 98 MDR-TB patients voluntarily isolated in urban hospitals in Nigeria reported an average hospital stay of 4.5 ± 1.9 months, with isolation ending after 2 negative sputum results [33].

PICO 1e: Effect of Respiratory Isolation Compared to No Isolation on Patient and/or Health System Costs

Patient costs due to isolation were grouped into loss of income, food insecurity, transportation costs, housing insecurity, and impacts on education. Surveys of Nigerian MDR-TB patients reported concerns about income (65/98, 66%) and job loss (50/98, 51%) [33]. Interviews with Canadian immigrants with smear-positive pulmonary TB (typically isolated for 3 weeks) described negative impacts on employment and finances, which improved but did not abate months after diagnosis [40]. A qualitative study from Canada reported isolation poses particular challenges for people with severe, persistent mental illness and recommended social protection policies [35]. Interviews with adolescents with drug-susceptible TB, caregivers, and health workers in Peru described negative perceived and experienced effects on current and future educational prospects, including scholarships, career trajectory, and/or employability [20].

A mixed-methods study in San Diego, California, United States, demonstrated potential cost savings associated with isolation of 7 homeless people with smear-positive pulmonary TB in supervised community housing programs (cost of \$56/day) compared to hospital isolation (cost of \$651/day) with projected total cost savings of \$189 242 (\$27 035 per patient) [34]. A descriptive study of voluntarily isolated Nigerian MDR-TB patients highlighted prohibitive costs of hospital admissions and concerns about increased community transmission due to long waiting lists [33].

PICO 1f: Effect of Respiratory Isolation Compared to No Isolation on Patient Mental Health or Stigma

In a retrospective, population-based cohort in Canada, depression risk was 24% higher in immigrants with TB than without TB (adjusted hazard ratio [aHR] 1.24, 95% confidence interval [CI]: 1.12–1.31) and length of hospital stay (known to be prolonged for TB patients and considered as a surrogate for

isolation) mediated 50% of depression risk [36]. Mixed methods studies from the United States and Nigeria indicated hospital isolation led to mood disturbance, poor psychological well-being, and social disconnection due to impacts on family, social, or economic activities [33, 37]. Qualitative studies highlighted the negative psychological impacts of isolation, compounded in populations with existing mental illnesses and facing intersectional stigma and overlapping health disparities, including migrants/immigrants, indigenous people and incarcerated people [20, 21, 35, 38–41]. The stress of isolation, particularly if prolonged, included family disruption and increased care needs [20, 21, 38, 40]. Transmission fears, including concerns about reinfection or acquiring other infections, in hospital or prison settings, were raised [39, 41]. Although stigma was not measured, fear of transmission, which underlies isolation policies, reinforced discrimination experienced by adolescents with TB in Peru and caused social exclusion affecting disclosure decisions in immigrants with TB in Canada [20, 40].

PICO 2a and 2b: Effect of Respiratory Isolation Compared to Masking on TB Infection and Disease in Contacts

A quasi-experimental, controlled trial in Chile assigned patients initiating treatment for pulmonary TB to either receive a bundle of respiratory prevention measures, including index patient separation and masking, or routine care. Preliminary data showed no difference in TB infection in household contacts between the intervention and control groups (8.5% vs 8.1%, RR = 1.05, 95% CI: .38–2.86) [43]. A modeling study evaluating different IPC interventions on XDR-TB incidence in the hospital and community catchment area in South Africa suggested patients with XDR-TB wearing surgical masks alone could reduce overall XDR-TB incident cases by 5%, whereas a combined strategy, including shorter hospital stays, more outpatient treatment, and patient and staff respiratory mask use, could avert 33% of cases [31].

PICO 2c–2f: No studies had data on the effect of isolation compared to masking on mortality, duration of hospitalization, patient or health system costs, and mental health or stigma.

Quality Assessment

Included studies were largely of low quality (Supplementary Table 2), primarily due to gaps in design or outcome reporting related to the effects of isolation, indirectness (since isolation was not the sole intervention), unaccounted for confounders in non-randomized studies, non-response bias in quantitative studies, and concerns about interpreting results related to isolation not being sufficiently substantiated by data in qualitative studies. Due to the limited data available, no studies were excluded based on quality. Overall study quality was reflected in our certainty of evidence assessment (Table 3), where the strength of evidence was consistently downgraded due to concerns about bias, indirectness, and inconsistency.

Table 3. Certainty of Evidence Assessment for PICO Questions 1 and 2

PICO Question	Domains Assessed	Certainty of Evidence Assessment	Notes
PICO 1a.: Does isolation reduce TB infection among contacts?	Study design	Very low	A single RCT; 3 modeling studies. Uncertain effect size. Downgraded due to limitations in study design, quality, inconsistency of findings across studies, and indirectness and applicability of studied interventions.
PICO 1b: Does isolation reduce TB incidence among contacts?	quality assessment	Very low	
PICO 1c: Does isolation reduce TB mortality?	inconsistency Indirectness	Very low	
PICO 1d: Does isolation increase duration of hospitalization for PWTB?		Moderate	Primarily observational studies. Downgraded for study design, and indirectness to current guideline intervention (community based isolation) with most studies evaluating hospital based isolation duration. Effect size was consistent across studies suggesting increased duration with respiratory isolation.
PICO 1e: Does isolation increase patient and/or health system costs?		Moderate	Primarily observational studies. Downgraded for study design, and indirectness to current guideline intervention (community based isolation) with several studies evaluating hospital based isolation, with prolonged durations. Effect size was consistent across studies suggesting increased health system costs with respiratory isolation.
PICO 1f: Does isolation worsen mental health outcomes and stigma among PWTB?		Moderate	Primarily observational and qualitative studies. Downgraded for study design, and indirectness to current guideline intervention (community based isolation) with several studies evaluating hospital based isolation, and prolonged durations. Effect size was consistent across studies suggesting worse psychological well-being and mental health and increased stigma associated with respiratory isolation.

Abbreviations: PICO, populations, interventions, comparators, and outcomes; PWTB, people with tuberculosis; RCT, randomized controlled trial; TB, tuberculosis.

DISCUSSION

Limited Data to Suggest Effectiveness of Isolation for Reducing TB Infection and Disease Incidence; Isolation Rarely Implemented Without Other Interventions

This is the first systematic review to our knowledge to evaluate the effects of respiratory isolation to reduce community-based TB transmission on public health and patient-important outcomes. Seventeen studies in low, medium, and high TB incidence countries were included. One RCT directly compared an isolation and non-isolation intervention and did not find a difference in the rates of TB infection or disease in household contacts [27, 28, 42]. Limitations include high baseline contact TB infection rates, reflecting high baseline community transmission, which may have obscured differences between study arms, concerns that contacts who remained uninfected at the time of index patient treatment initiation may have also been less susceptible to infection, and low numbers of contacts who developed TB disease or who died [44]. Although modeling studies suggested isolation may reduce incident TB infection and disease, uncertainties including the duration of infectiousness after treatment initiation limit the assumptions underpinning the models [29–31]. Many studies reported the negative effects of isolation on individuals and communities, including duration of hospitalization, costs to patients and health systems, including impacts on employment and education, and effects on mental health impact and stigma, often driven by fears about infectiousness [20, 21, 33, 35, 37–40, 45]. The lack of studies directly comparing isolation with no isolation or masking, along with heterogeneity in study design,

approaches to outcome measurement and reporting, led to judgements of overall low study quality and high concerns for risk of bias and indirectness.

Evidence in Context

Another systematic review that informed the latest WHO IPC guidelines on facility-based TB transmission, found few studies, often of low quality, and highlighted the lack of evidence on isolation and difficulty disaggregating the effects of simultaneously implemented IPC measures [15]. A recent review informing Canadian Thoracic Society de-isolation guidelines also concluded isolation after appropriate treatment initiation may have little impact on reducing TB transmission and has a net negative impact on patient and societal well-being [46].

Gaps in the Literature

There was a lack of studies directly assessing the impact of home or community isolation, which was our intended focus, though some effects of hospital isolation (when used to reduce community transmission) may be extrapolated to home isolation. Details about the form or duration of isolation, criteria for de-isolation, or whether masking was implemented, were often absent. Isolation practices were often heterogeneous and variably implemented with limited data to assess implementation [47] or effects on outcomes of interest. Effective treatment is the key intervention to stop transmission [14], yet most included studies lacked details on when treatment was started or likelihood of treatment effectiveness.

Isolation is more likely to be effective in the pre-treatment period (or pre-effective treatment period, in the case of drug resistance).

Ethical Implications of Isolation and Critical Need for Adequate Support for PWTB

Respiratory isolation, particularly when prolonged and/or facility-based, raises ethical concerns due to curtailment of individual liberties and human rights for the benefit of the community [19]. Included studies emphasized the disproportionate negative economic and psychosocial effects of isolation on marginalized populations, including immigrants, Indigenous people, and incarcerated people. Health systems imposing isolation should mitigate impacts on health equity through a multisectoral approach that could include job security, food vouchers, transportation and peer navigators. TB programs should acknowledge structural violence experienced by TB-affected communities [21, 38], including forced removals, the lack of culturally-competent, patient-centered care, and stigmatization on community re-entry [38]. Programs should examine reasons for home isolation not being offered to certain groups, and develop outreach programs with multilingual clinicians and support workers to provide culturally competent care [40].

Strengths and Limitations

Strengths of our review include a comprehensive search strategy developed with an expert librarian. All screening, abstraction, and quality assessment were undertaken in duplicate by a diverse team with medical and public health expertise. Limitations include not searching gray literature (materials that are unpublished or published outside the typical academic publishing environment) and the possibility of missing studies excluded due to language. Data heterogeneity and weaknesses in study designs, including the lack of studies evaluating isolation as the sole intervention, precluded meta-analyses, which may have been of value to policymakers. Insufficient studies also precluded stratified analyses to examine the effects on prevention of drug-susceptible versus drug-resistant TB transmission.

Recommendations for Future Research

Most de-isolation recommendations remain contingent on smear conversion, which is a poor predictor for infectiousness post-treatment initiation [48]. Newer diagnostic tools, including devices or masks to capture cough aerosols or *Mtb* DNA, may help to identify more precisely when patient infectiousness ceases [49–53]. Given the challenges in designing trials to measure the effects of isolation, implementation research to monitor and evaluate how isolation is applied, and its impacts on communities, can help to address some of these data gaps.

Conclusions

There are limited data to suggest isolation reduces community-based TB transmission, particularly when patients are on

confirmed or presumed effective treatment. Multiple studies identified negative effects of isolation, including costs, mental health, and stigma, which disproportionately impact marginalized groups. Included studies were overall of low quality, and evidence for the effect of isolation was largely indirect, limiting utility. Policymakers must weigh the negative impacts of isolation on people with TB against the community benefit of reduced transmission and consider a tailored approach for isolation that is evidence-based, equitable, and aligned with person-centered TB care.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

Acknowledgments. The authors wish to thank Paul Bain at the Harvard Countway Library of Medicine, who helped design and conduct the searches for this systematic review.

Financial support. R. R. N. is supported by a National Institutes of Health (NIH) Patient-Oriented Mentored Career Development Award (award number NIAID K23 AI132648-06) and NIH Director's New Innovator Award (award number NIAID DP2-AI176896).

Potential conflicts of interest. The authors: No reported conflicts of interest. All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

References

1. World Health Organization. 2023. Global Tuberculosis Report 2022. World Health Organization; Available at: <https://apps.who.int/iris/handle/10665/363752>. Accessed 5 September 2024.
2. Dye C. Evolution of tuberculosis control and prospects for reducing tuberculosis incidence, prevalence, and deaths globally. *JAMA* 2005; 293:2767.
3. Styblo K. The relationship between the risk of tuberculous infection and the risk of developing infectious tuberculosis. *Bull Int Union Tuberc* 1985; 60:117–9.
4. van Geuns H, Meijer J, Styblo K. Results of contact examination in Rotterdam, 1967–1969. *Bull Int Union Tuberc* 1975; 50:107–21.
5. Dharmadhikari AS, Mphahlele M, Venter K, et al. Rapid impact of effective treatment on transmission of multidrug-resistant tuberculosis. *Int J Tuberc Lung Dis* 2014; 18:1019–25.
6. Riley RL, Mills CC, O'Grady F, Sultan LU, Wittstadt F, Shivpuri DN. Infectiousness of air from a tuberculosis ward. *Am Rev Respir Dis* 1961; 85: 511–25.
7. Shaikh A, Sriraman K, Vaswani S, Oswal V, Rao S, Mistry N. Early phase of effective treatment induces distinct transcriptional changes in *Mycobacterium tuberculosis* expelled by pulmonary tuberculosis patients. *Sci Rep* 2021; 11:17812.
8. Riley RL, Moodie AS. Infectivity of patients with pulmonary tuberculosis in inner city homes. *Am Rev Respir Dis* 1974; 110:810–2.
9. Walter ND, Dolganov GM, Garcia BJ, et al. Transcriptional adaptation of drug-tolerant *Mycobacterium tuberculosis* during treatment of human tuberculosis. *J Infect Dis* 2015; 212:990–8.
10. Walter ND, Born SEM, Robertson GT, et al. *Mycobacterium tuberculosis* precursor rRNA as a measure of treatment-shortening activity of drugs and regimens. *Nat Commun* 2021; 12:2899.
11. Jindani A, Aber VR, Edwards EA, Mitchison DA. The early bactericidal activity of drugs in patients with pulmonary tuberculosis. *Am Rev Respir Dis* 1980; 121: 939–49.
12. Jindani A, Doré CJ, Mitchison DA. Bactericidal and sterilizing activities of anti-tuberculosis drugs during the first 14 days. *Am J Respir Crit Care Med* 2003; 167:1348–54.

13. Nardell E, Stoltz A, de Kock E, et al. Measuring the early impact on transmission of new treatment regimens for drug resistant tuberculosis. *Am J Respir Crit Care Med* **2019**; 199:A7387.
14. Migliori GB, Nardell E, Yedilbayev A, et al. Reducing tuberculosis transmission: a consensus document from the World Health Organization Regional Office for Europe. *Eur Respir J* **2019**; 53:1900391.
15. Karat AS, Gregg M, Barton HE, et al. Evidence for the use of triage, respiratory isolation, and effective treatment to reduce the transmission of *Mycobacterium tuberculosis* in healthcare settings: a systematic review. *Clin Infect Dis* **2020**; 72: 155–72.
16. World Health Organization. WHO Guidelines on tuberculosis infection prevention and control: 2019 update. Geneva, Switzerland: World Health Organization, **2019**.
17. Jensen PA, Lambert LA, Iademaro MF, Ridzon R. Guidelines for preventing the transmission of *Mycobacterium tuberculosis* in health-care settings, 2005. *MMWR Recomm Rep* **2005**; 54:1–141.
18. Rouillon A, Perdriest S, Parrot R. Transmission of tubercle bacilli: the effects of chemotherapy. *Tubercle* **1976**; 57:275–99.
19. Mehra C, Lokhande D, Chavan D, Rane S. What quality of care means to tuberculosis survivors. *J Clin Tuberc Other Mycobact Dis* **2020**; 19:100157.
20. Oliva Rapoport VE, Altamirano E, Senador L, et al. Impact of prolonged isolation on adolescents with drug-susceptible tuberculosis in Lima, Peru: a qualitative study. *BMJ Open* **2022**; 12:e063287.
21. Mayan M, Robinson T, Gokiart R, Tremblay M, Abonyi S, Long R. Respiratory isolation for tuberculosis: the experience of Indigenous peoples on the Canadian prairies. *Public Health Action* **2017**; 7:275–81.
22. Shah M, Dansky Z, Nathavitharana R, et al. NTCA guidelines for respiratory isolation and restrictions to reduce transmission of pulmonary tuberculosis in community settings [manuscript published online ahead of print 18 April 2024]. *Clin Infect Dis* **2024**; ciae199. doi:10.1093/cid/ciae199
23. Covidence systematic review software. Available at: www.covidence.org. Accessed 10 September 2024.
24. The World Bank Group. What is an “international dollar”? World Bank Data Help Desk. Available at: <https://datahelpdesk.worldbank.org/knowledgebase/articles/114944-what-is-an-international-dollar>. Accessed 10 September 2024.
25. Hong QN, Fàbregues S, Bartlett G, et al. The Mixed Methods Appraisal Tool (MMAT) version 2018 for information professionals and researchers. *Education for Information* **2018**; 34:285–91.
26. Aromataris E, Munn Z, eds. JBI Manual for evidence synthesis. JBI, **2020**. Available at: <https://synthesismanual.jbi.global>. Accessed 10 September 2024.
27. Andrews RH, Devadatta S, Fox W, Radhakrishna S, Ramakrishnan CV, Velu S. Prevalence of Tuberculosis among close family contacts of tuberculous patients in South India, and influence of segregation of the patient on the early attack rate. *Bull World Health Organ* **1960**; 23:463–510.
28. Kamat SR, Dawson JJY, Devadatta S, et al. A controlled study of the influence of segregation of tuberculous patients for one year on the attack rate of tuberculosis in a 5-year period in close family contacts in South India. *Bull World Health Organ* **1966**; 34:517–32.
29. Selmane S. The impact of isolation of identified active tuberculosis cases on the number of latently infected individuals. In: *Computational Science and Its Applications—ICCSA 2011. Lecture Notes in Computer Science*, vol 6786. Berlin: Springer-Verlag, 2011:527–36.
30. Espindola AL, Varughese M, Laskowski M, Shoukat A, Heffernan JM, Moghadas SM. Strategies for halting the rise of multidrug resistant TB epidemics: assessing the effect of early case detection and isolation. *Int Health* **2017**; 9:80–90.
31. Basu S, Andrews JR, Poolman EM, et al. Prevention of nosocomial transmission of extensively drug-resistant tuberculosis in rural South African district hospitals: an epidemiological modelling study. *Lancet* **2007**; 370:1500–7.
32. Adler J, Smirnoff M, Klein S, et al. Hospital care of tuberculosis patients referred for directly observed therapy in New York City: identifying factors in prolonged stays. *J Public Health Manag Pract* **1995**; 1:22–7.
33. Oladimeji O, Ushie BA, Udoh EE, et al. Psychosocial wellbeing of patients with multidrug resistant tuberculosis voluntarily confined to long-term hospitalisation in Nigeria. *BMJ Glob Health* **2016**; 1:e000006.
34. LoBue PA, Cass R, Lobo D, Moser K, Catanzaro A. Development of housing programs to aid in the treatment of tuberculosis in homeless individuals: a pilot study. *Chest* **1999**; 115:218–23.
35. Silva DS, Komparic A, Upshur REG, Gibson J, Strike C. Provider perspectives on liberty and harm in the treatment of persons with tuberculosis and mental illness. *Int J Tuberc Lung Dis* **2019**; 23:405–11.
36. Basham CA, Karim ME, Cook VJ, Patrick DM, Johnston JC. Tuberculosis-associated depression: a population-based cohort study of people immigrating to British Columbia, Canada, 1985–2015. *Ann Epidemiol* **2021**; 63:7–14.
37. Kelly-Rossini L, Perlman DC, Mason DJ. The experience of respiratory isolation for HIV-infected persons with tuberculosis. *J Assoc Nurses AIDS Care* **1996**; 7: 29–36.
38. Moffatt J, Mayan M, Long R. Sanatoriums and the Canadian colonial legacy: the untold experiences of tuberculosis treatment. *Qual Health Res* **2013**; 23:1591–9.
39. Almeida Rueda LM, Giraldo Pineda A, Forero Pulido C. “Encierro sobre el encierro” aislamiento por tuberculosis. Centro penitenciario Medellín Colombia. *Salud UIS* **2016**; 48:61–9.
40. Bedingfield N, Lashewicz B, Fisher D, King-Shier K. Systems of support for foreign-born TB patients and their family members. *Public Health Action* **2022**; 12:79–84.
41. Horter S, Stringer B, Reynolds L, et al. “Home is where the patient is”: a qualitative analysis of a patient-centred model of care for multi-drug resistant tuberculosis. *BMC Health Serv Res* **2014**; 14:81.
42. Tuberculosis Chemotherapy Centre. A concurrent comparison of home and sanatorium treatment of pulmonary tuberculosis in South India. *Tubercle* **1959**; 40: 468–76.
43. Balcells ME, Ruiz-Tagle C, Seguel R, et al. ECCMID, Barcelona, Spain. **2023**.
44. Menzies D. Effect of treatment on contagiousness of patients with active pulmonary tuberculosis. *Infect Control Hosp Epidemiol* **1997**; 18:582–6.
45. Bendjelloul G, Gerard S, Birgand G, et al. Impact of admission to high-risk isolation room on patients’ and healthcare workers’ perceptions: a qualitative cross-assessment approach. *Infect Dis Now* **2021**; 51:247–52.
46. Cooper R. Appendix B: de-isolation review and recommendations. *Can J Respir Crit Care Sleep Med* **2022**; 6:248–55.
47. Salerno M, Burzynski J, Behm H. A survey of Tuberculosis home isolation practices in the United States. In: Presented at: NTCA Annual Conference. Atlanta, GA. **2023**.
48. Cooper R, Williams M, Fennelly KP. Abandon the acid-fast bacilli smear for patients with TB on effective treatment. *Chest* **2023**; 164:21–3.
49. Acuña-Villaorduña C, Schmidt-Castellani LG, Marques-Rodrigues P, et al. Cough-aerosol cultures of *Mycobacterium tuberculosis* in the prediction of outcomes after exposure. A household contact study in Brazil. *PLoS One* **2018**; 13: e0206384.
50. Jones-López EC, Namugga O, Mumbowa F, et al. Cough aerosols of *Mycobacterium tuberculosis* predict new infection: a household contact study. *Am J Respir Crit Care Med* **2013**; 187:1007–15.
51. Theron G, Limberis J, Venter R, et al. Bacterial and host determinants of cough aerosol culture positivity in patients with drug-resistant versus drug-susceptible tuberculosis. *Nat Med* **2020**; 26:1435–43.
52. Williams CM. Exhaled *Mycobacterium tuberculosis* predicts incident infection in household contacts. *Clin Infect Dis* **2023**; 76:e957–64.
53. Williams CM, Abdulwhhab M, Birring SS, et al. Exhaled *Mycobacterium tuberculosis* output and detection of subclinical disease by face-mask sampling: prospective observational studies. *Lancet Infect Dis* **2020**; 20:607–17.